METRIC

MIL-PRF-8565/4D
w/AMENDMENT 2
01 November 2021
SUPERSEDING
MIL-PRF-8565/4D (AS)
w/AMENDMENT 1
19 July 2012

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MAINTENANCE-FREE, 24-VOLT, 7.5-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

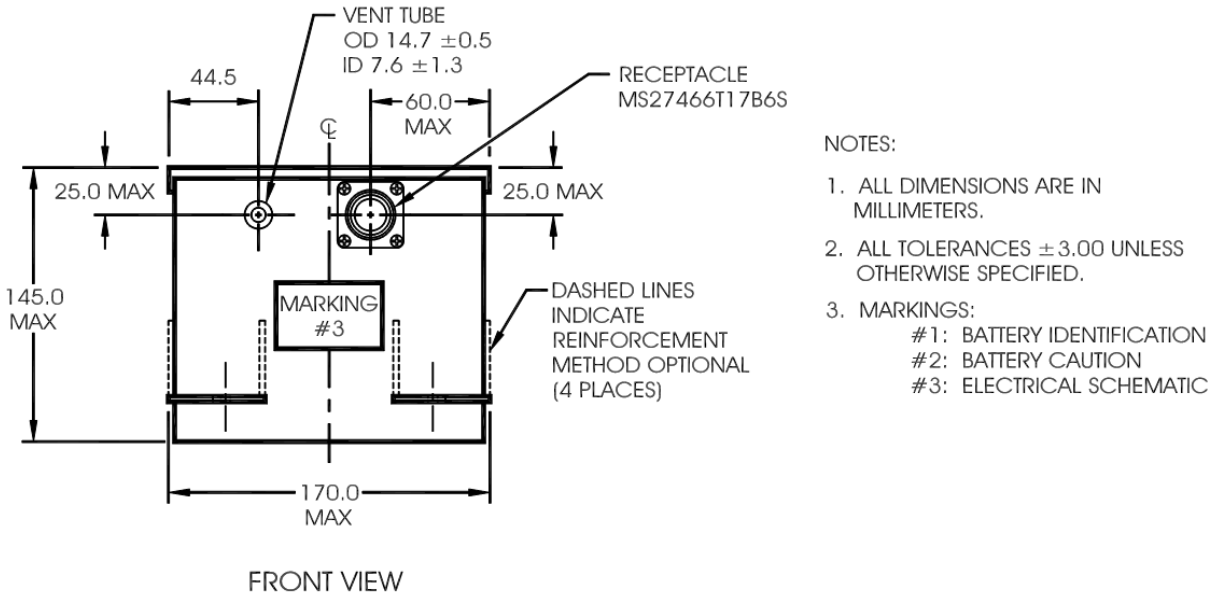


FIGURE 1. Dimensions and configuration for PIN D8565/4-1.

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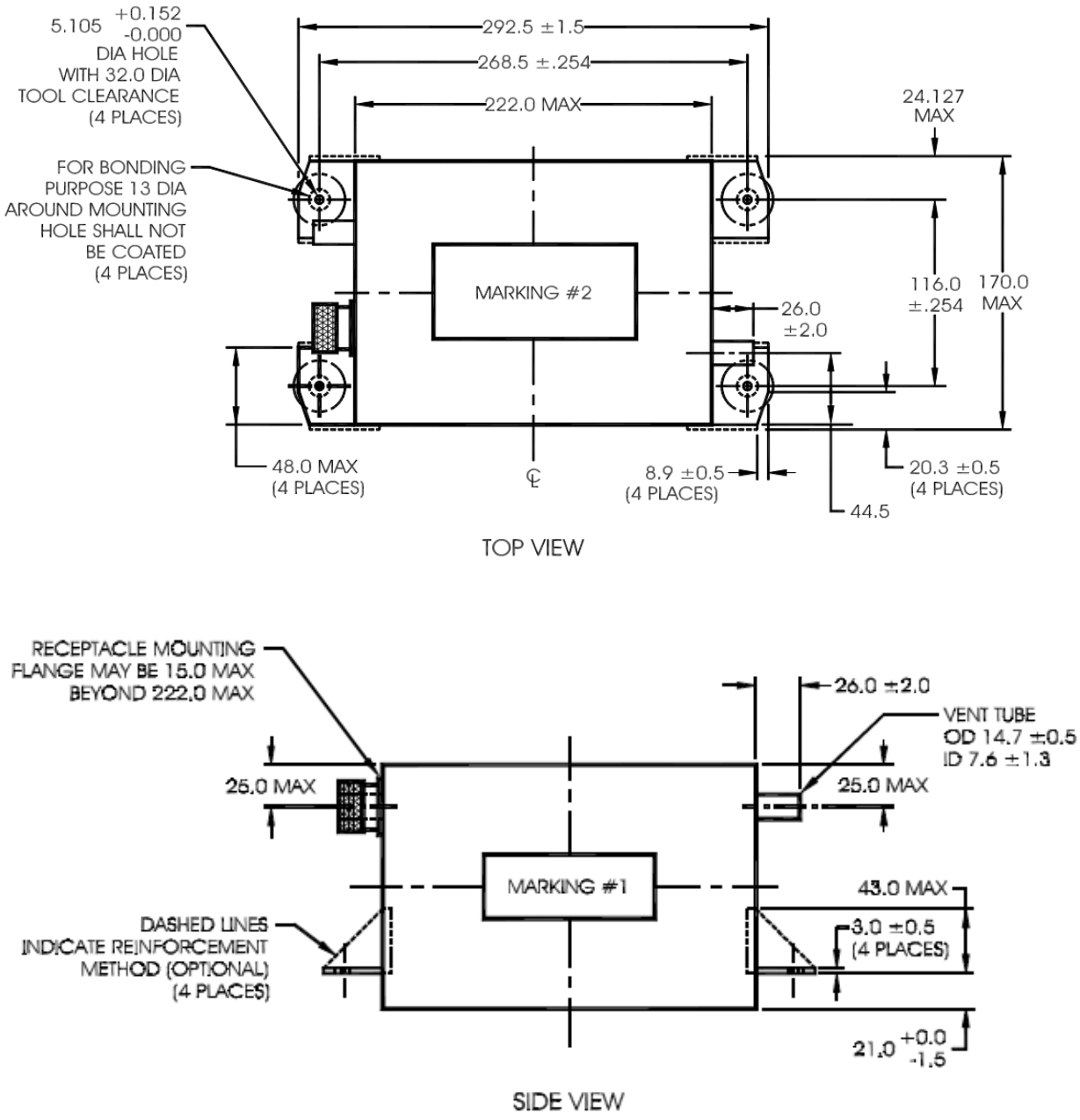


FIGURE 1. Dimensions and configuration for PIN D8565/4-1 – Continued.

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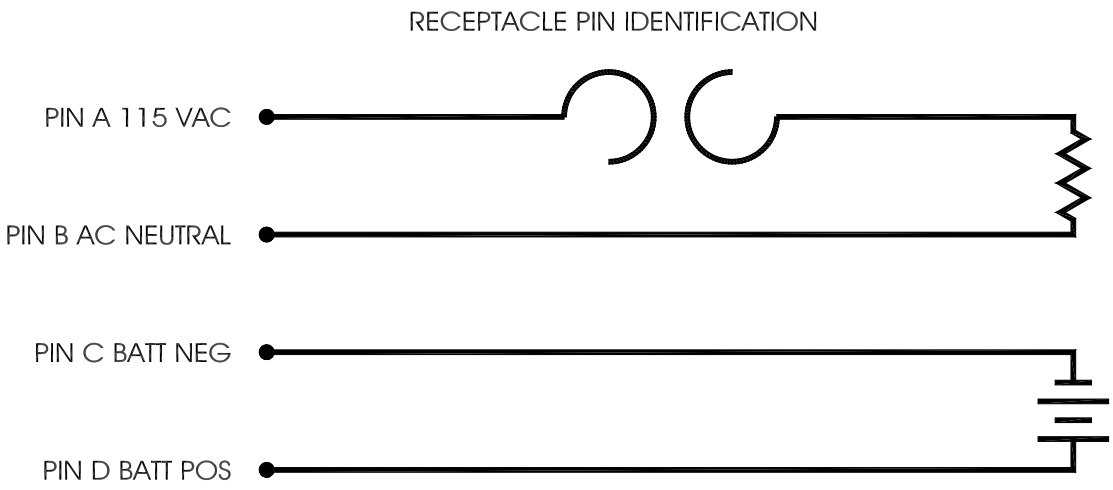


FIGURE 2. Electrical schematic for marking #3.

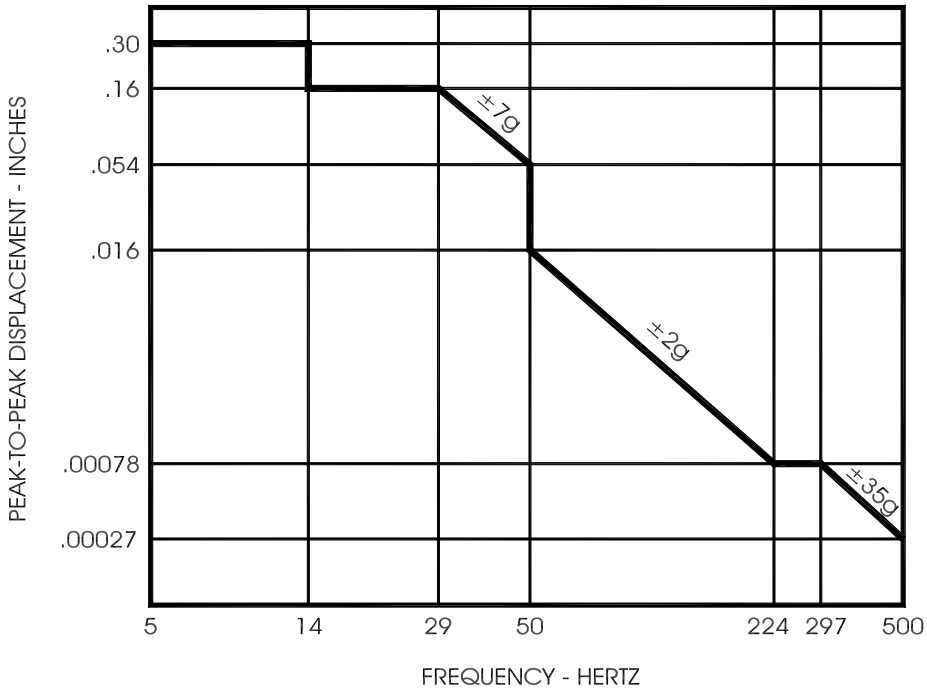


FIGURE 3. Sine wave vibration test levels.

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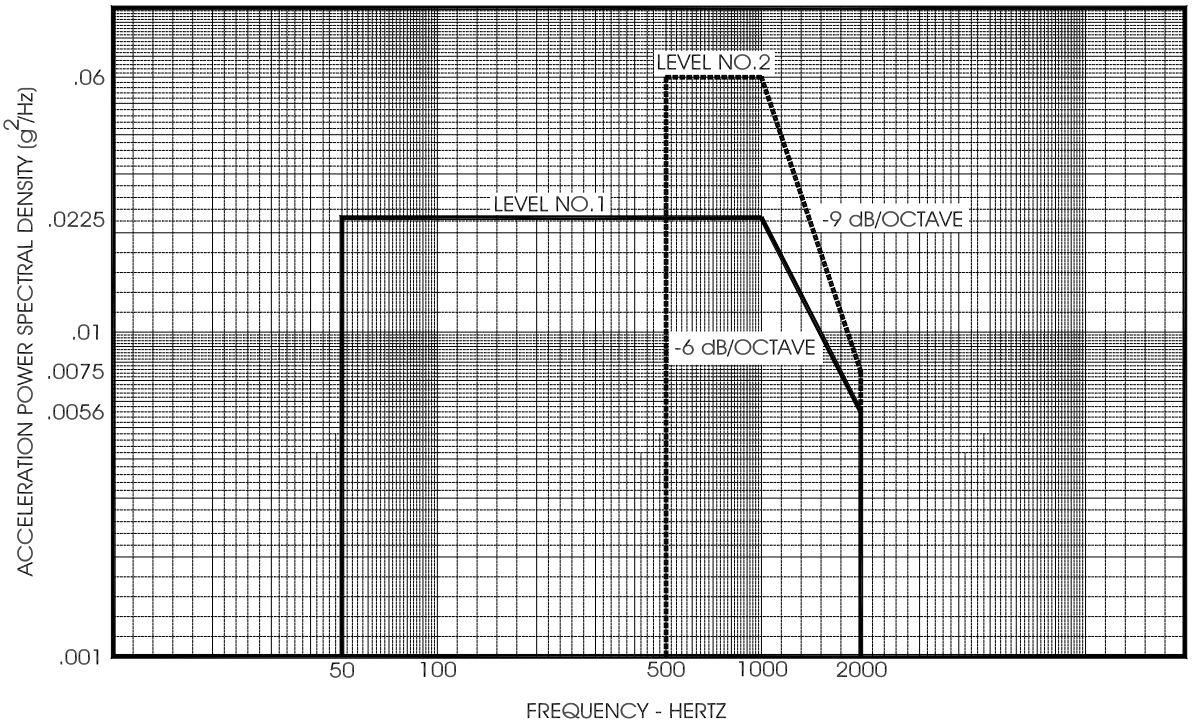
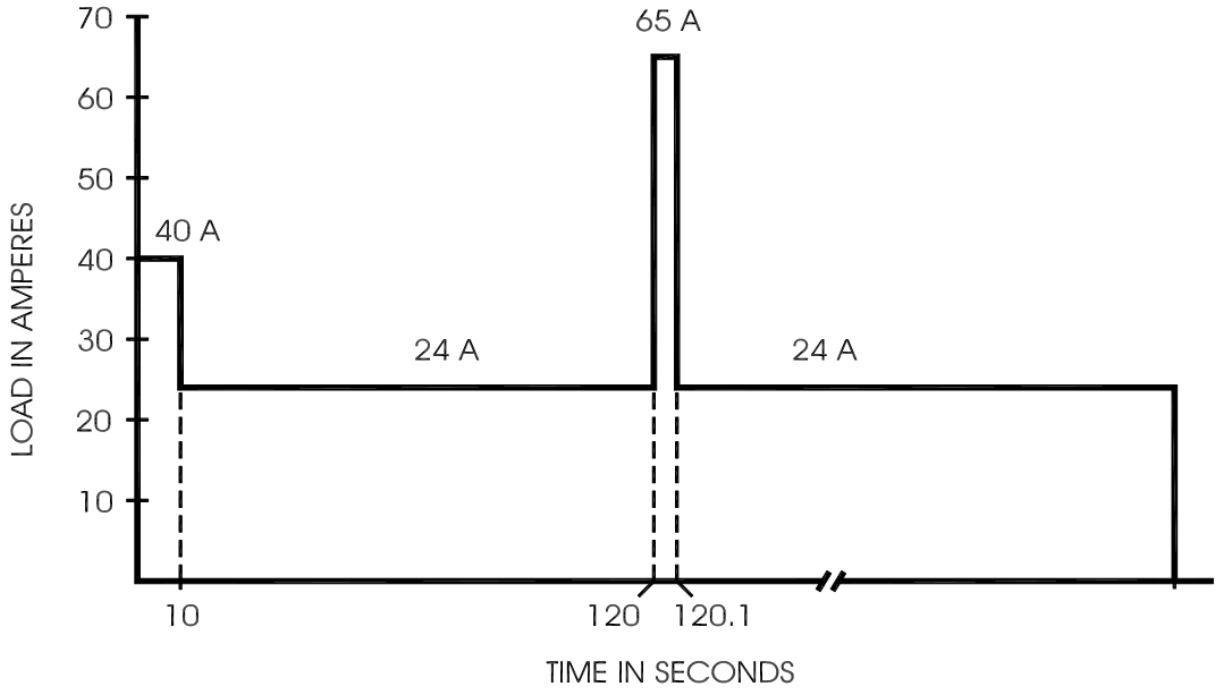


FIGURE 4. Random vibration test levels.

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Step	Sample number	Discharge temperature (°C)	Load current (amperes)	Discharge time (seconds)
1	1, 2, 3	25	40 24 65 24	10 110 0.1 480
2	1	-29	40 24 65 24	10 110 0.1 50
3	2	-18	40 24 65 24	10 110 0.1 180
4	3	0	40 24 65 24	10 110 0.1 300
5	1	55	40 24 65 24	10 110 0.1 480

FIGURE 5. Aircraft emergency load requirements.

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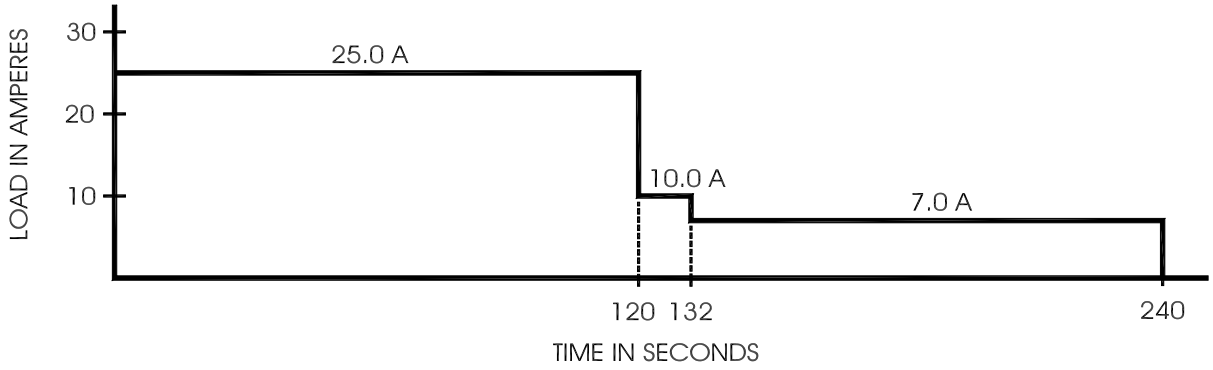


FIGURE 6. Aircraft start-up load requirements.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be D8565/4-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 contained in the specification sheet. The weight of the battery shall be not greater than 11.8 kilograms (see 3.8.3 and 4.5.3).

3. Capacity. The rated capacity of the battery shall be not less than 7.5 ampere-hours (Ah). See 3.8.6 as modified herein.

4. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown on figure 3 of MIL-PRF-8565. Vibration isolation shall be provided to ensure that the vibration level does not exceed the levels of the 2-year accelerated life test of 4.5.10. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided.

5. Qualification inspection and conformance inspection. Table I lists the required inspections.

6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

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6.1 The following paragraphs and figure are not applicable:

3.6.2 Polarity marking.
3.8.7 Constant voltage discharge.
3.8.8 Strength of receptacle.
4.5.8 Constant voltage discharge.
4.5.9 Strength of receptacle test.
4.5.12 Charge and discharge test at low temperature.
4.5.30 Wormgear retaining pin hardness.

6.2 Modify the following paragraphs:

3.5.5 Receptacles. Insert the following after the first sentence: "The battery's electrical interface to the aircraft shall be the MS27466T17B6S receptacle. It shall be positioned and electrically connected as shown on figures 1 and 2, respectively, of the specification sheet.

"a. Pin A shall be high side AC (input heater power).

"b. Pin B shall be AC ground (heater return).

"c. Pin C shall be battery negative.

"d. Pin D shall be battery positive."

3.5.10 Heaters. In 3.5.10.b, delete the text and insert the following: "Heater current shall be not greater than 3.0 amperes. The rated current of each heater control device shall be compatible with the heater current."

3.6.5 Marking for batteries with heaters. In the first sentence, delete "battery identification marking and battery caution marking" and insert "battery cover and the surface bearing the battery identification". After the third sentence, insert: "(see figure 2 of the specification sheet)."

3.8.6 Capacity and electrical performance. Delete the text and insert: "The rated capacity of the battery shall be not less than 7.5 ampere-hours. To meet the capacity requirement, each battery shall produce a potential of not less than 18.0 volts at not less than 60 minutes when connected to a 7.5-ampere load after being tested in accordance with 4.5.7 at 21°C (70°F) (7.5 Ah/1.0 hr./21°C/18.0 V at end of life). The battery's 1.0 C-rate shall be 7.5 amperes (see requirement 5 of the specification sheet)."

3.8.16 Output performance. Delete the text and insert: "The battery shall produce a potential of not less than 18.0 volts during and after the tests of 4.5.28."

3.8.17 Cold and hot temperature charge. Delete the text.

6.3 Add the following paragraphs:

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"3.8.17.1 Half-hour charging. The battery shall produce not less than 3.5 ampere-hours after being tested in accordance with 4.5.29.1.c. The battery shall produce not less than 7.0 ampere-hours after being tested in accordance with 4.5.29.1.h. The battery shall also meet the requirements of 3.7.

"3.8.17.2 One-hour charging. The battery shall produce not less than 7.0 ampere-hours after being tested in accordance with 4.5.29.2.e. The battery shall not sustain any damage when a 28.25 ± 0.5 volt charging source is applied continuously to the battery during the test of 4.5.29.2. The battery shall also meet the requirements of 3.7."

6.4 Modify the following paragraphs:

4.5.19 Vibration test. Delete 4.5.19.a and 4.5.19.d and insert the following: "d. Energize the battery heaters during vibration. Vibrate the battery along each orthogonal axis (in the axis order shown on figure 3 of the general specification) in accordance with the test levels on figure 3 of the specification sheet. Subject the battery to two 15-minute logarithmic sweeps from 5-500-5 Hertz. Conduct one 5-minute logarithmic sweep from 20-30-20 Hertz. Conduct a resonance search from 5-500 Hertz. Perform resonance dwell testing for 1.5 minutes at each resonance, to a maximum of three most severe exhibiting a transmissibility of two or greater. Vibrate the battery for a period of 20 minutes per axis in accordance with level 1 on figure 4 of the specification sheet. Vibrate the battery for a period of 5 minutes per axis to level 2 on figure 4 of the specification sheet." Delete the last sentence of 4.5.19.e and insert the following: "Discharge the battery at 25.0 amperes for 2 minutes before beginning each axis."

4.5.28 Output performance test. Delete the text and insert the following: "Each of the three battery samples shall be subjected to each of the following tests."

6.5 Add the following paragraphs:

"4.5.28.1 Aircraft emergency loads. This test consists of the following steps.

"a. Discharge each sample in accordance with step 1 of figure 5 of the specification sheet.

"b. Constant potential charge each sample at room temperature in accordance with 4.4.2 (see .4.1).

"c. Stabilize each sample battery's temperature at the test temperature specified below (see .4.1). With each sample still in the temperature chamber at the indicated test temperature, discharge each sample as follows:

"(1) Sample 1 in accordance with step 2 of figure 5 of the specification sheet.
"(2) Sample 2 in accordance with step 3 of figure 5 of the specification sheet.
"(3) Sample 3 in accordance with step 4 of figure 5 of the specification sheet."

"d. Constant potential charge each sample at room temperature in accordance with 4.4.2 (see 4.4.1).

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"e. Discharge sample 1 in accordance with step 5 of figure 5 of the specification sheet.

"f. Constant potential charge sample 1 at room temperature in accordance with 4.4.2 (see 4.4.1).

"g. Examine each sample for the requirements of 3.8.16.

"4.5.28.2 Aircraft start-up loads. This test consists of the following steps.

"a. Stabilize the temperature of sample 1 at 55°C (131°F), sample 2 at –29°C (-20°F), and sample 3 at 25°C (77°F) for 12 hours.

"b. With each battery still in the temperature chamber at the indicated test temperature, discharge each sample in accordance with figure 6 of the specification sheet.

"c. Examine each sample for the requirements of 3.8.16."

6.6 Modify the following paragraph:

4.5.29 Cold and hot temperature charge test. Delete the text.

6.7 Add the following paragraphs:

"4.5.29.1 Half-hour charge test. This test consists of the following steps.

"a. Constant potential charge the battery in accordance with 4.4.2 for 5 hours at room temperature (see 4.4.1).

"b. Stabilize the samples (see 4.4.1) at - 18°C (- 0°F) for sample 1, 15°C (59°F) for sample 2, and 55°C (131°F) for sample 3. Perform 4.5.29.1.c through 4.5.29.1.g with the battery in the environmental chamber at the specified temperature.

"c. Discharge each sample at the 1.0 C-rate to a cutoff voltage of 18.0 volts.

"d. Examine the battery for the requirements of 3.8.17.1.

"e. Stabilize the samples at 25°C (77°F) (see 4.4.1). Constant potential charge the battery in accordance with 4.4.2.

"f. Stabilize the samples (see 4.4.1) at the temperatures of 4.5.29.1.b. Discharge each sample at the 1.0 C-rate to remove all but 3.5 ampere-hours from each sample as determined by 4.5.29.1.d.

"g. Stabilize the samples (see 4.4.1) at the temperatures of 4.5.29.1.b. Constant potential charge the battery in accordance with 4.4.2 for 30 minutes.

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“h. Immediately after completing 4.5.29.1.g, place the samples in a room temperature environment (see 4.4.1). Discharge the samples at the 1.0 C-rate to a cutoff voltage of 18.0 volts.

“i. Examine the battery for the requirements of 3.8.17.1.

“4.5.29.2 One-hour charge test. This test consists of the following steps.

“a. Constant potential charge the battery in accordance with 4.4.2.

“b. Stabilize the battery (see 4.4.1) at - 40°C (- 40°F). Perform 4.5.29.2.c and 4.5.29.1.d with the battery in the environmental chamber at the specified temperature.

“c. Discharge the battery at the 1.0 C-rate to a cutoff voltage of 18.0 volts.

“d. Stabilize the battery (see 4.4.1) at - 40°C (- 40°F). Constant potential charge the battery in accordance with 4.4.2 for 1.0 hour.

“e. Immediately after completing 4.5.29.2.d, place the battery in a room temperature environment (see 4.4.1) and discharge at the 1.0 C-rate to a cutoff voltage of 18.0 volts.

“f. Examine the battery for the requirements of 3.8.17.2.”

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1	4.5.1		X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	3.5, 3.6, 3.8.2, table II	4.5.2	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2, 3.8.3	4.5.3	1/	X	
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	3.7, 3.8.4	4.5.4	1/	X	X
5	STRENGTH OF VENT TUBES TEST	X	X	X	3.7, 3.8.5	4.5.5	1/	X	X
6	ELECTRICAL TEST PREPARATION	X	X	X		4.5.6		X	X
7	CAPACITY DISCHARGE	X	X	X	Reqmt 3, 3.8.6	4.5.7	X	X	X
8	AIRCRAFT EMERGENCY LOADS	X	X	X	3.8.16	4.5.28.1			
9	AIRCRAFT START-UP LOADS	X	X	X	3.8.16	4.5.28.2			
10	HALF-HOUR CHARGE TEST	X	X	X	3.7, 3.8.17.1	4.5.29.1			
11	ONE-HOUR CHARGE TEST	X	X	X	3.7, 3.8.17.2	4.5.29.2			
12	CYCLING TEST	X			3.8.9, reqmt (1) of table III	4.5.10		X	
13	CAPACITY DISCHARGE AT 49°C (120°F)		X		3.8.6, reqmt (2) of table III	4.5.11			
14	DISCHARGE WHILE INVERTED		X		3.5.9, reqmt (5) of table III	4.5.13			X
15	ALTITUDE TEST		X		3.7, 3.5.9, 3.8.10	4.5.14			X
16	SHOCK TEST (BASIC DESIGN)				3.7, 3.8.10	4.5.15		X	
17	TEMPERATURE SHOCK TEST	X			3.7, 3.8.10	4.5.16			

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TABLE I. Qualification inspection and conformance inspection of batteries – Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph	Test paragraph	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
18	TEMPERATURE RISE AND FLOAT TEST		X		3.7, 3.8.11, reqmt (1) of table III	4.5.17			X
19	BATTERY GAS EMISSION TEST		X		3.8.10, 3.18.12	4.5.18			
20	HUMIDITY TEST		X		3.7, 3.8.10	4.5.20			
21	VIBRATION TEST		X		3.7, 3.8.10	4.5.19			
22	SALT FOG TEST	X			3.7, 3.8.10	4.5.21			
23	GROUND STORAGE TEST			X 2/	3.8.13.1	4.5.22			
24	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7, 3.8.10, reqmt (1) of table III	4.5.23			
25	DEEP DISCHARGE RECOVERY TEST			X	3.7, 3.8.13.3	4.5.27			
26	SHOCK TEST (CRASH SAFETY)			X	3.8.10, 3.8.15	4.5.25			
27	IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER)	3/	3/	3/	3.8.14	4.5.24			
28	SHELF LIFE TEST			X	3.8.9, 3.8.13.2	4.5.26			
29	FINAL EXAMINATION	X	X	X	3.7, 3.8.4	4.5.31		X	X

1/ See sample information in test paragraph.
2/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.
3/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.

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CONCLUDING MATERIAL

Custodian:
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Air Force - 170
DLA - CC

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Navy - AS

(Project 6140-2021-019)

Review activity:
Air Force - 70

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